

Novel Detection and Progression Markers for Diabetes Based on Continuous Glucose Monitoring Data Dynamics

Eslam Montaser,¹ Leon S. Farhy,^{2,3} and Boris P. Kovatchev²

¹Division of Endocrinology and Metabolism, Department of Medicine, Indiana University School of Medicine, Indianapolis, IN 46202, USA

²Center for Diabetes Technology, School of Medicine, University of Virginia, Charlottesville, VA 22903, USA

³Division of Endocrinology and Metabolism, Department of Medicine, School of Medicine, University of Virginia, Charlottesville, VA 22903, USA

Correspondence: Eslam Montaser, PhD, Division of Endocrinology and Metabolism, Department of Medicine, Indiana University School of Medicine, 1120 W Michigan St, Gatch Hall 380D, Indianapolis, IN 46202, USA. Email: emontase@iu.edu.

Abstract

Context: Static measures of continuous glucose monitoring (CGM) data, such as time spent in specific glucose ranges (70–180 mg/dL or 70–140 mg/dL), do not fully capture the dynamic nature of blood glucose, particularly the subtle gradual deterioration of glycemic control over time in individuals with early-stage type 1 diabetes.

Objective: Develop a diabetes diagnostic tool based on 2 markers of CGM dynamics: CGM entropy rate (ER) and Poincaré plot (PP) ellipse area (S).

Methods: A total of 5754 daily CGM profiles from 843 individuals with type 1, type 2 diabetes, or healthy individuals with or without islet autoantibody status were used to compute 2 individual dynamic markers: ER (in bits per transition; BPT) of daily probability matrices describing CGM transitions between 8 glycemic states, and the area S (mg²/dL²) of individual CGM PP ellipses using standard PP descriptors. The Youden index was used to determine “optimal” cut-points for ER and S for health vs diabetes (case 1); type 1 vs type 2 (case 2); and low vs high type 1 immunological risk (case 3). The markers’ discriminative power was assessed through the area under the receiver operating characteristics curves (AUC).

Results: Optimal cutoff points were determined for ER and S for each of the 3 cases. ER and S discriminated case 1 with AUC = 0.98 (95% CI, 0.97–0.99) and AUC = 0.99 (95% CI, 0.99–1.00), respectively (cutoffs ER_{case1} = 0.76 BPT, S_{case1} = 1993.91 mg²/dL²), case 2 with AUC = 0.81 (95% CI, 0.77–0.84) and AUC = 0.76 (95% CI, 0.72–0.81), respectively (ER_{case2} = 1.00 BPT, S_{case2} = 5112.98 mg²/dL²), and case 3 with AUC = 0.72 (95% CI, 0.58–0.86), and AUC = 0.66 (95% CI, 0.47–0.86), respectively (ER_{case3} = 0.52 BPT, S_{case3} = 923.65 mg²/dL²).

Conclusion: CGM dynamics markers can be an alternative to fasting plasma glucose or glucose tolerance testing to identify individuals at higher immunological risk of progressing to type 1 diabetes.

Key Words: continuous glucose monitoring, entropy rate, islet autoantibodies, Poincaré plot, stages of diabetes, transition probability matrix

Abbreviations: 2hPG, 2-hour plasma glucose; Ab, autoantibody; AUC, area under the receiver operating characteristics curves; BPT, bits per transition; CGM, continuous glucose monitoring; CLC, closed-loop control; CV, coefficient of variation; DCLP1, The International Diabetes Closed Loop (iDCL) Trial: Protocol 1 (iDCL) study; DCLP3, The International Diabetes Closed Loop (iDCL) Trial: Clinical Acceptance of the Artificial Pancreas (DCLP3) study; DCLP5, The International Diabetes Closed Loop (iDCL) Trial: Clinical Acceptance of the Artificial Pancreas in Pediatrics: A Study of t:Slm X2 With Control-IQ Technology; DIA1, Multiple Daily Injections and Continuous Glucose Monitoring in Diabetes (DIA1) study; DIA2, Multiple Daily Injections and Continuous Glucose Monitoring in Diabetes (DIA2) study; ER, continuous glucose monitoring entropy rate; FPG, fasting plasma glucose; HbA_{1c}, glycated hemoglobin A_{1c}; MDI, multiple daily injections; NoNDiab, Continuous Glucose Monitoring Profiles in Healthy Nondiabetic Participants: A Multicenter Prospective Study; MG, mean continuous glucose monitoring; OGTT, oral glucose tolerance test; PP, Poincaré plot; ROC, receiver operating characteristic; S, ellipse area; SAP, sensor-augmented pump; SD1, ellipse width; SD2, ellipse length; TN, Insulin-Glucose-Glucagon Network: Defining a Type 1 Diabetes Progression Index study; WTP, weighted average transition probability matrix.

Diabetes presents a global health challenge, and it is critical to be diagnosed or predicted as early as possible to prevent or delay long-term health complications, improve quality of life, avoid premature death, and reduce distress (1, 2).

The American Diabetes Association, the International Expert Committee, and the World Health Organization have recommended different diagnostic criteria for diabetes. These criteria are based either on fasting plasma glucose (FPG) levels, on the 2-hour plasma glucose (2hPG) level after a 75-g oral glucose tolerance test (OGTT), or on glycated

hemoglobin A_{1c} (HbA_{1c}) level, with corresponding cutoff points for diagnosing of diabetes (3–6).

As a test for the initial diagnosis of diabetes, FPG is easier than the 2hPG to perform at the hospital but requires fasting overnight before a blood draw too, which is a time-consuming process. However, the 2hPG is more sensitive than FPG alone and can capture the dynamic of glucose spikes after the meal but is limited by poor reproducibility in the 60% to 80% range (7–9). Besides that, the macronutrient content of the meal before the OGTT may affect the results (10). In addition,

Received: 11 March 2024. Editorial Decision: 28 May 2024. Corrected and Typeset: 5 July 2024

© The Author(s) 2024. Published by Oxford University Press on behalf of the Endocrine Society. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com. See the journal About page for additional terms.

OGTT-defined categories have limitations, especially when one considers the transition from dysglycemia to the onset of the disease (11). The HbA_{1c} is an accurate and reproducible test that has the advantage of convenience compared with FPG or 2hPG, because it does not require fasting overnight. However, HbA_{1c} may be an unreliable test under certain conditions such as hemolytic anemia, iron deficiency, hemoglobinopathies, and pregnancy (12). Furthermore, racial differences in HbA_{1c} independent of blood glucose have been reported in many different studies (13–15).

In the last 2 decades, continuous glucose monitoring (CGM) systems have proved the ability to track dynamic glycemic fluctuations, trends, and patterns over time, allowing for the optimization of medical therapy, minimizing the risk of hypoglycemia, improving quality of life, screening, or diagnosing diabetes (16, 17). Several studies have demonstrated that CGM-derived time in range (70–180 mg/dL) has a high correlation with HbA_{1c} and can be used as a marker to predict diabetes-related complications (18–20). Over the years, a number of CGM-derived glycemic metrics have been developed to assess the quality of glycemic control through aggregating CGM data to communicate a meaningful clinical message (21). Therefore, CGM offers superior reliability and clinical usefulness over traditional methods like FPG, 2hPG, or HbA_{1c}, albeit at a higher cost, providing continuous real-time glucose monitoring, comprehensive insights into glycemic trends, and personalized diabetes management strategies. However, representing CGM data through the ambulatory glucose profile and CGM-derived time in different ranges does not reflect the progression of glycemic control over time. Since the essential advantage of CGM is measuring time series of glucose values and capturing the process of glycemic control as it evolves, adding a temporal component to CGM-derived times in range is now possible.

Recently, in several studies in adolescents and children, CGM data were used for detecting early hyperglycemia and to characterize participants who progressed to stage 3 type 1 diabetes with respect to the percentage of time they spent above 140 mg/dL (22–26). Other studies have shown that integrating CGM data into machine-learning models could help develop predictive models to assist clinicians in improving the screening, diagnosing, and treatment of diabetes (27). Classifying and clustering daily CGM profiles into one of 32 sets of clinically similar clusters have been proposed to allow tracking of daily glycemic changes over time (28).

Given the limitations of the current diagnostic tests for diabetes, we propose new dynamical approaches that include 2 different methods. The first method is based on the entropy of the dynamical system representing diabetes, and includes (i) encoding CGM data into strings of letters, each letter identifying a specific glycemic state; (ii) computing a transition probability matrix between sequential daily glycemic states; (iii) computing the proportion of time an individual spends in each glycemic state, that is, the stationary distribution of the transition probability matrix, and (iv) computing the entropy rate of the encoded CGM process using the transition probability matrix and stationary distribution for each individual. This method is based on the theory of semi-Markov chains, used here to describe the evolution of an individual across a predefined set of glycemic states, characterized by random duration of time spent in each state. The second method is based on visualizing and quantifying the attractor of the dynamical system representing diabetes. It includes (i) representing CGM data through a Poincaré plot (PP) as a simple visual technique for analyzing

dynamical systems, and (ii) computing the area of the fitting ellipse (S) through the descriptors of PP geometry.

Materials and Methods

Data Overview

The data used in this work were collected across 7 different studies (Table 1) that tested various treatment modalities in individuals with type 1 diabetes (DCLP1; NCT02985866 (29), DCLP3; NCT03563313 (30), DCLP5; NCT03844789 (31), and DIA1; NCT02282397 (32) studies), type 2 diabetes (DIA2; NCT02282397 (33) study), or healthy individuals without islet autoantibody (Ab) status (NoNDiab; Type 1 diabetes Exchange Clinic Network Studies (34)) and with a different Ab status (TN; NCT02663661 (35) study—Negative Ab, 1 Ab, ≥ 2 Ab). CGM data were collected for the entire duration of the studies. All CGM data sets were collected using versions of the Dexcom sensor as follows: Dexcom G4/G5 was used in DCLP1, Dexcom G6 in DCLP3, DCLP5, and NoNDiab, and Dexcom G4 Platinum was used in DIA1, DIA2, and TN.

Only complete daily CGM profiles (a time series of 288 glucose data points collected every 5 minutes during the midnight-to-midnight [24-hour] period) from the 7 aforementioned studies were used in the analysis. Individuals with fewer than 5 CGM profiles completed in a week were excluded from the analysis.

Encoding the Continuous Glucose Monitoring Data Space

To describe the CGM transitions through different glycemic states, the daily CGM profiles are encoded as strings of letters as follows; further details are given in Supplement 1 (36):

- (i) First, the dimensionality of the 288-point daily CGM profiles was reduced by splitting it into 96 fifteen-minute segments, each consisting of 3 adjacent CGM measurements.
- (ii) The mean CGM (MG) is computed for each segment.
- (iii) Each 3-point segment is then assigned a letter (glycemic state) according to its MG value:

a if MG is less than 54 mg/dL, *b* if 54 is less than or equal to MG less than 69 mg/dL, *c* if 70 is less than or equal to MG less than 120 mg/dL, *d* if 121 is less than or equal to MG less than 140 mg/dL, *e* if 141 is less than or equal to MG less than 160 mg/dL, *f* if 161 is less than or equal to MG less than 180 mg/dL, *g* if 181 is less than or equal to MG less than 250 mg/dL, and *h* if MG is greater than 250 mg/dL.

Transition Probability Matrix for Representation of Daily Continuous Glucose Monitoring Profiles

Transition or change in the daily CGM traces from one state (letter) to another in the next period of time (eg, transition from time in range to time above range), is a random process and is expressed in the form of probability, and it is called the “transition probability,” used to determine the probability of the next state (37). Each daily CGM profile can be presented by a transition probability matrix $P = (P_{ij})$ of size 8×8 such that the individual entries per individual are the probability of moving from state i to state j . For every i and j , the

Table 1. Clinical and demographic characteristics of the 7 different data sets used for analysis

Study	DCLP1	DCLP3	DCLP5	DIA1	DIA2	TN	NoNDiab
Data type	Type 1 diabetes	Type 1 diabetes	Type 1 diabetes	Type 1 diabetes	Type 2 diabetes	Healthy	Healthy
Treatment modality	CLC, SAP	CLC, SAP	CLC, SAP	MDI	MDI	Non	Non
No. of participants (experimental group)	125 (64)	168 (112)	101 (77)	158 (105)	158 (79)	60	163
Average age, y	32.5	33	11	48	60	23.7	31
Sex, % female	47	50	49	44	56	58.3	66
Race, % White	87	89	81	NR	63	100	93
Average HbA _{1c} %, (mmol/mol)	7.4 (57)	7.4 (57)	7.8 (62)	8.6 (70)	8.5 (69)	5.3 (34)	5.1 (32)

Abbreviations: CLC, closed-loop control; DCLP1, The International Diabetes Closed Loop (iDCL) Trial: Protocol 1 (iDCL) study; DCLP3, The International Diabetes Closed Loop (iDCL) Trial: Clinical Acceptance of the Artificial Pancreas (DCLP3) study; DCLP5, The International Diabetes Closed Loop (iDCL) Trial: Clinical Acceptance of the Artificial Pancreas in Pediatrics: A Study of t:Slm X2 With Control-IQ Technology; DIA1, Multiple Daily Injections and Continuous Glucose Monitoring in Diabetes (DIA1) study; DIA2, Multiple Daily Injections and Continuous Glucose Monitoring in Diabetes (DIA2) study; HbA_{1c}, glycated hemoglobin A_{1c}; MDI, multiple daily injections; NoNDiab, Continuous Glucose Monitoring Profiles in Healthy Nondiabetic Participants: A Multicenter Prospective Study; NR, not reported; SAP, sensor-augmented pump; TN, Insulin-Glucose-Glucagon Network: Defining a Type 1 Diabetes Progression Index study.

P matrix should satisfy the following conditions: $P_{ij} \geq 0$, $\forall i, j$, and $\sum_{j=1}^n P_{ij} = 1$, $\forall i$. Eight different glycemic states $\{a, b, c, d, e, f, g, h\}$ were used as described earlier to define the P matrix for each daily CGM profile per individual across the different studies. Besides that, the weighted average transition probability matrix (WTP) was computed for n daily CGM traces per individual through the number of transitions from one state to another. Finally, the stationary distribution, D , of the WTP matrix is computed. Of note, each individual has one D , representing the proportion of time spent in each of the 8 glycemic states.

Entropy Rate as a Biomarker

The entropy rate (ER) is a measure of the uncertainty and complexity of daily CGM traces. ER marker is measured in bits per transition (BPT). A smaller value of the ER corresponds to a more predictable and less complex daily CGM profile (37). Therefore, a low ER transition matrix represents a daily CGM profile for an individual with a higher probability of transitions to the same states (eg, spending most time in, say, state c), and vice versa. ER is computed from the WTP matrix and its corresponding stationary distribution D per individual (I) as follows:

$$ER(I) = - \sum_{i,j} D_i WTP_{ij} \log_2 WTP_{ij}.$$

Describing the Dynamics of Continuous Glucose Monitoring Through Poincaré Plot

The PP, or lag plot, was developed by the French mathematician Henri Poincaré in 1890, as a simple technique for visualization of the attractor of a dynamical system. In essence, the plot is a 2-dimensional projection of the phase space trajectory and, in diabetes, visualizes the variability in the dynamics of the CGM time series (38). In the PP, the plot appears as a stretched cloud of points oriented along the line of identity; each point of the plot has coordinates $G(t)$ on the x-axis and $G(t+\Delta t)$ on the y-axis, where $G(t)$ is the glucose CGM value at time t and $G(t+\Delta t)$ is the glucose CGM value at time $t+\Delta t$. Thus, the difference ($y-x$) in coordinates of each data point represents the glucose rate of change occurring between times t and $t+\Delta t$. We constructed PPs with fixed $\Delta t = 15$ minutes, based on the assumption that a lag period of 15 minutes is optimal for the

evaluation of the blood glucose rate of change (39); however, other values are possible, ranging from 5 minutes to an hour.

Quantitative Descriptors of Poincaré Plot Geometry

A PP trajectory can be characterized by several descriptors; SD1 and SD2 are 2 standard PP descriptors (40). SD1 (the width of the ellipse) is the SD of the projection of the PP on the line perpendicular to the line of identity ($y = -x$; reflects the short-term variability), and SD2 (the length of the ellipse) is defined as the SD of the projection of the PP on the line of identity ($y = x$; reflecting the long-term variability). Both descriptors are defined as follows:

$$SD1^2 = \frac{1}{2} SD\Delta G^2,$$

$$SD2^2 = 2SD^2 - SD1^2.$$

$SD\Delta G$ is the SD of the successive ($G_{t+\Delta t} - G_t$) intervals, and SD is the SD of the glucose levels. In our analysis, we use SD1 and SD2 to compute the area of the fitting ellipse (S) which reflects the total variability per individual (I) as follows

$$S(I) = \pi SD1(I)SD2(I).$$

Statistical Analysis

All statistical analyses were performed using R Statistical Software 4.0.5 (R Foundation for Statistical Computing). Continuous variables were described as mean, and categorical variables were summarized as frequency (percentage). The area under the receiver operating characteristics (ROC) curves (AUC) was calculated to evaluate the discriminative power of the dynamic markers to distinguish between healthy and individuals with diabetes (case 1), distinguish between type 1 diabetes and type 2 diabetes individuals (case 2), and distinguish between individuals with low risk and high risk of developing type 1 diabetes (case 3). The “optimal” cut-points of the ER and S in the 3 different cases of diagnosis were estimated by the maximum value of the Youden index. The Pearson correlation matrix between the ER, S , the proportion of time spent in the 8 glycemic states (letters), and coefficient of variation (CV) was computed, to assess the collinearity between glycemic metrics and both dynamic markers (ER and S).

Results

In total, the CGM time series data collected across all 7 studies described earlier contained CGM traces for 924 individuals and 55 655 complete daily CGM profiles. Following CGM data preprocessing, a total of 81 individuals were excluded, as they did not have valid daily CGM profiles (ie, 55 individuals from NoNDiab, 21 individuals from TN [6 from the Negative group, 7 from the 1 Ab group, and 8 from the ≥ 2 Ab group], 3 individuals from DIA2, and 2 individuals from the DCLP1 study). By comparing CGM traces with different lengths, we established that a 1-week complete CGM profile is sufficient for the purpose of this analysis. Therefore, 5754 complete daily CGM profiles from the remaining 843 individuals were used in the analysis, with a maximum of 1-week CGM data per individual. Most (65.6%) of the daily CGM profiles were generated by individuals with type 1 diabetes treated by multiple daily insulin injections (MDI), a sensor-augmented pump (SAP), or closed-loop control (CLC) system. The DIA2 study had individuals with type 2 diabetes on MDI treatment and representing 18.8% of the daily CGM profiles. The NoNDiab study had healthy individuals without information about their Ab status and represented 11.8% of the daily CGM profiles. The TN study had also healthy individuals with known Ab status: 15 participants with zero/Negative Ab, 11 participants with 1 Ab, and 13 participants with 2 or more Abs. They represented 3.8% of the daily CGM profiles. In addition, one study focused on children with type 1 diabetes (DCLP5). The data set for the analysis was therefore a mix of individuals with type 1 and type 2 diabetes of different ages, following different treatments, and healthy individuals, some of whom have known low or high immunological risk of developing type 1 diabetes.

Tracking the Progression of Disease Through a Transition Probability Matrix

In healthy studies (TN and NoNDiab), all individuals never visit the glycemic state *b* and have a high probability of staying

in the glycemic state *c* (*c*, 70–120 mg/dL). In addition, all individuals with Negative Abs in the TN study never visit states *b* or *g*, unlike some individuals with 2 or more Abs, who can visit state *g* with high probability. Therefore, the WTP matrix can detect dysglycemia for those individuals with 2 or more Abs (at risk of developing type 1 diabetes) at an early stage. In studies of individuals with diabetes (type 1 and type 2 of diabetes); all individuals can visit states *b* and *g*, normally as hyperglycemic glycemic states, but in varying proportions based on the treatment.

The proportion of time the individual spends in each glycemic state was computed through the stationary distribution per study and is summarized in Supplementary Table S1 (36). The main difference between the healthy (with or without Ab status) and studies of individuals with diabetes (type 1 and type 2 of diabetes) is the proportion of time spent in the euglycemic state *c*, which decreased gradually, from the Negative Ab individuals (88.07%) to those individuals with 2 or more Abs (78.38%) in the TN study; this kind of transition could detect dysglycemia in those individuals. Those individuals in the Negative Ab, 1 Ab groups, and the NoNDiab study were spending the majority of time in state *c* (>83%). Therefore, we expect those individuals to spend most time in state *c* as the CGM time series evolves. In addition, those individuals with type 1 and type 2 diabetes were spending the majority of time in state *g* (24.31% and 28.24%, respectively). Hence, the transition probability matrix of the probabilities of moving from one glycemic state to another could be used as a tool to observe disease or treatment progression for each individual across the predefined set of glycemic states.

Entropy Rate and the Area of the Fitting Poincaré Ellipse as Dynamic Markers

Fig. 1 shows the average ER (BPT) and *S* (mg^2/dL^2) values with the SEM per study by using 1-week CGM data. Individuals with Negative Abs in the TN study had a minimum average ER (*S*) = 0.39 BPT (615.93 mg^2/dL^2), and

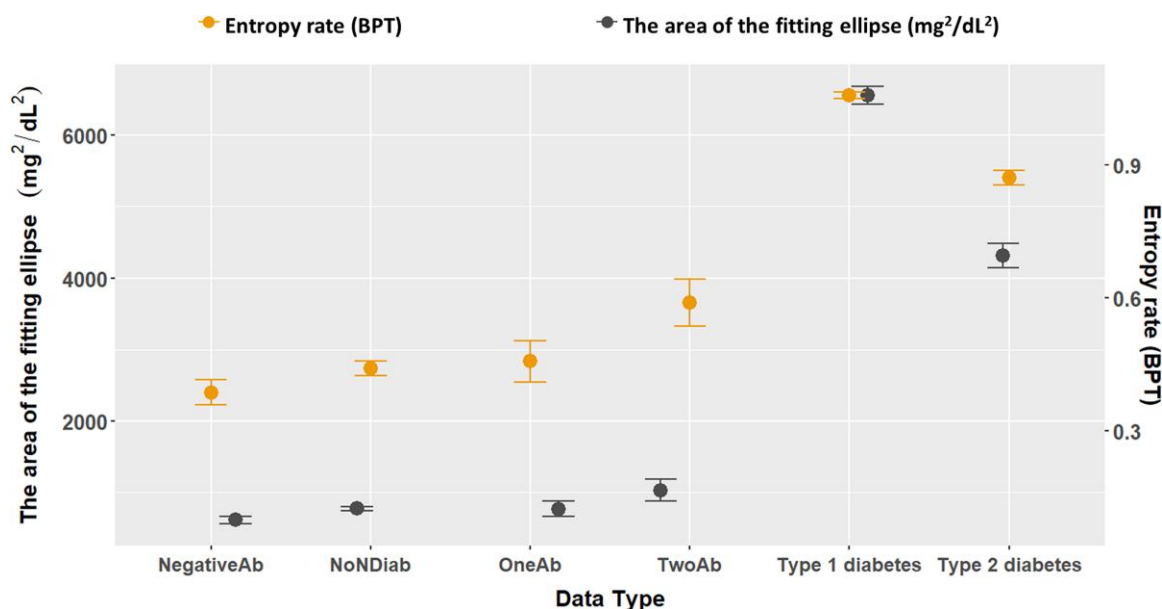


Figure 1. Scatterplots with the SE for the entropy rate criterion (bits per transition; BPT), and the area of the fitting ellipse (mg^2/dL^2) for the different data sets (TN study: NegativeAb, 1 Ab, and ≥ 2 Abs; NoNDiab study, Type 1 diabetes studies with different treatments, and Type 2 diabetes study with multiple daily insulin injections [DIA2]), through using 1-week continuous glucose monitoring data. Ab, islet autoantibody.

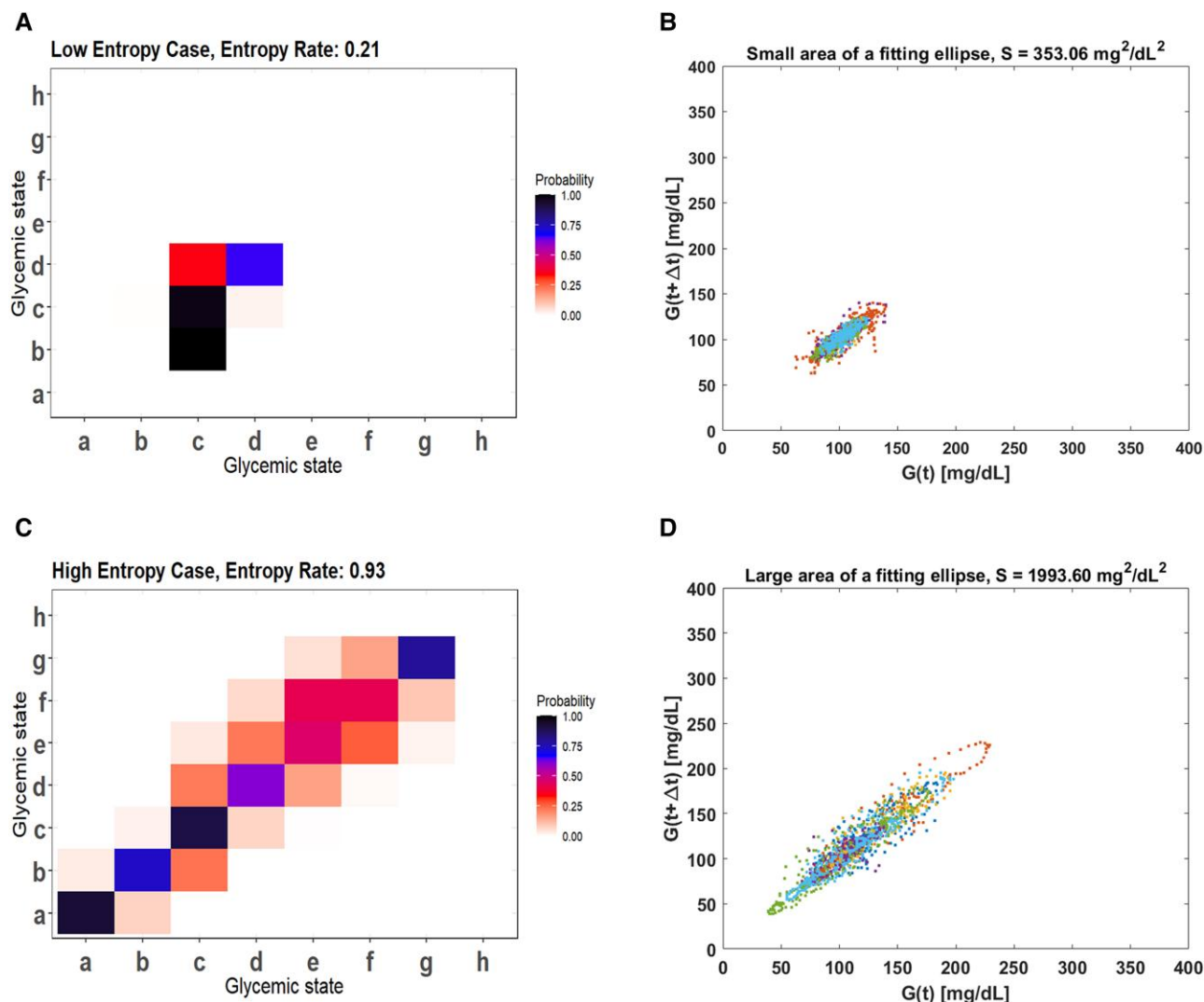


Figure 2. An illustrative example of the transition probability matrix and the corresponding area of Poincaré plot ellipse for 2 different individuals from the TN study (A and B, an autoantibody-negative individual vs C and D, an autoantibody-positive individual). A represents the weighted average transition probability matrix with a low entropy rate (entropy rate = 0.21 bits per transition [BPT]) for an autoantibody-negative individual, where the x-axis and y-axis represent the 8 glycemic states defined as (a if mean CGM glucose [MG] < 54 mg/dL, b if 54 ≤ MG < 69 mg/dL, c if 70 ≤ MG < 120 mg/dL, d if 121 ≤ MG < 140 mg/dL, e if 141 ≤ MG < 160 mg/dL, f if 161 ≤ MG < 180 mg/dL, g if 181 ≤ MG < 250 mg/dL, and h if MG > 250 mg/dL), and colored zones represents the probability of transition from glycemic state to another; all glycemia are in zones c and d (low-risk zones); B represents the corresponding small area of the Poincaré plot ellipse ($S = 353.06 \text{ mg}^2/\text{dL}^2$) with a $\Delta t = 15$ minutes; C represents the weighted average transition probability matrix with a high entropy rate (entropy rate = 0.93 BPT) for an autoantibody-positive individual (ie, an individual has ≥2 autoantibodies), and colored zones represent the probability of transition from glycemic state to another. Most glycemia are in zones a to e and progressed to zone f and g (high-risk zones). D represents the corresponding large area of the Poincaré plot ellipse ($S = 1993.60 \text{ mg}^2/\text{dL}^2$) with a $\Delta t = 15$ min. TN, Insulin-Glucose-Glucagon Network: Defining a Type 1 Diabetes Progression Index study.

individuals with type 1 diabetes had the highest average ER ($S = 1.06$ BPT ($6581.70 \text{ mg}^2/\text{dL}^2$)). The average ER (S) for individuals from healthy studies (TN and NoNDiab) was lower than for individuals from type 1 and type 2 diabetes studies. Those individuals with 2 or more Abs have an average ER = 0.59 BPT, and $S = 1032.75 \text{ mg}^2/\text{dL}^2$, higher than individuals with 1 Ab (ER = 0.46 BPT, $S = 769.39 \text{ mg}^2/\text{dL}^2$) and Negative Abs (ER = 0.39 BPT, $S = 615.93 \text{ mg}^2/\text{dL}^2$) and individuals from the NoNDiab study (ER = 0.45 BPT, $S = 776.10 \text{ mg}^2/\text{dL}^2$). In addition, the average ER (S) for individuals from the DIA2 study (type 2 diabetes with MDI), was lower than for individuals from type 1 diabetes studies, with ER = 0.87 BPT, and $S = 4313.01 \text{ mg}^2/\text{dL}^2$. Fig. 2A and 2B show an example of an individual from the Negative Ab group

with a low ER transition matrix and a small S vs another individual from the 2 or more Ab group with a high ER transition matrix and a large S (Fig. 2C and 2D). The Pearson correlation matrix was computed between the ER and the 8 glycemic states, CV(%), and S . The ER has a correlation with the 8 glycemic states, especially, with state c ($r = -0.65$), and state e ($r = 0.68$). As well, ER has a high correlation with CV ($r = 0.71$) and S ($r = 0.53$), but has a weak correlation with state a ($r = 0.15$) and state b ($r = 0.02$).

The AUC of the ROC curve was used to evaluate the discriminative power of both biomarkers for separating healthy from individuals with diabetes or type 1 from type 2 diabetes individuals, or low risk from high risk of developing type 1 diabetes. Fig. 3 shows the ROC curves for the 3 different cases

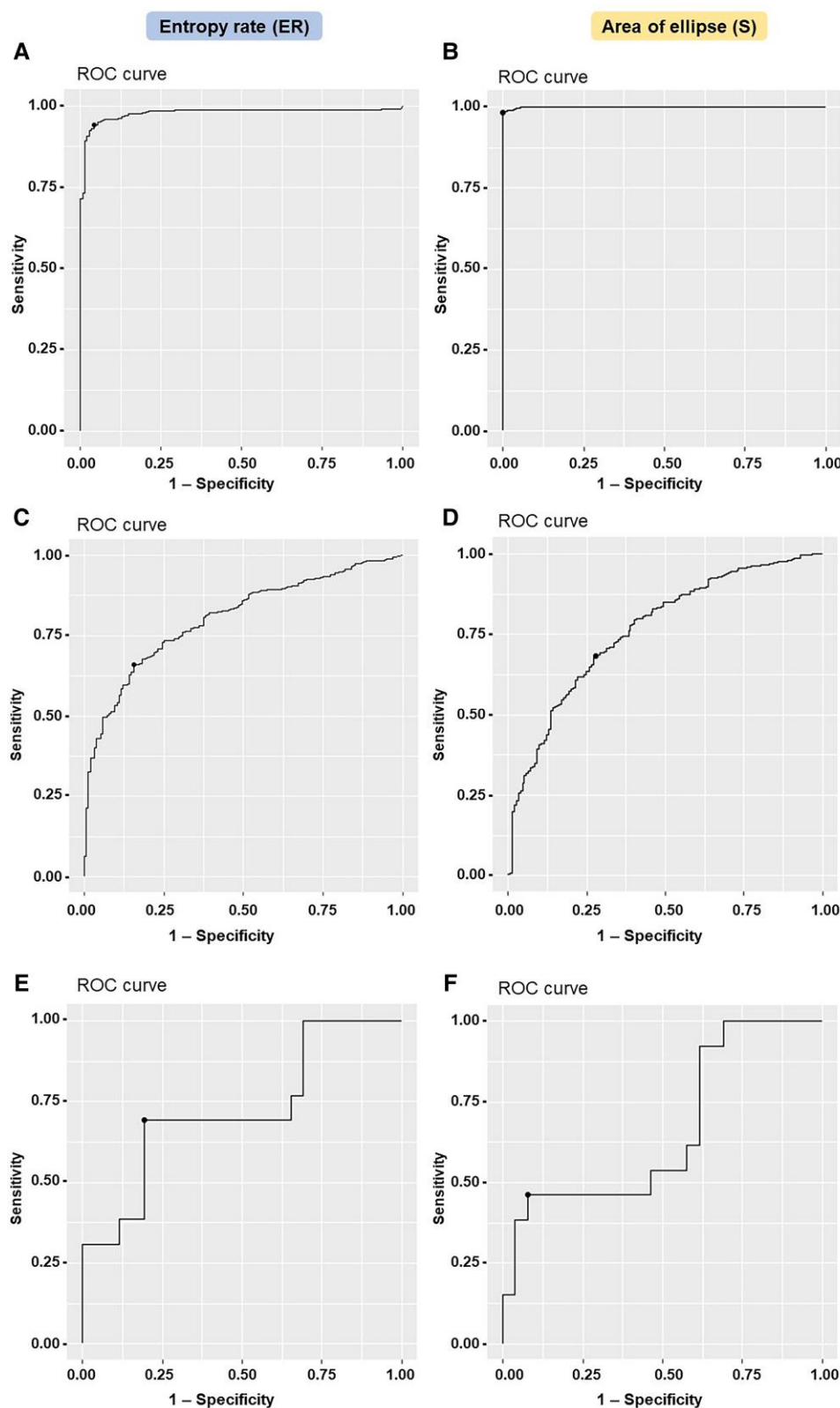


Figure 3. The receiver operating characteristic (ROC) curves for 3 different cases of discrimination to evaluate the discriminative power of the dynamic biomarkers (ie, entropy rate [ER] and area of Poincaré plot ellipse [S]). A and B, ROC curves for the discrimination between healthy and individuals with diabetes; C and D, ROC curves for the discrimination between type 1 diabetes and type 2 diabetes individuals; and E and F, ROC curves for the discrimination between individuals with low risk and high risk of developing type 1 diabetes.

of discrimination, where ER and S with an “optimal” cutoff point of 0.76 (BPT), and 1993.91 (mg^2/dL^2), respectively, were able to discriminate healthy vs individuals with diabetes, with an AUC of 0.98 (95% CI, 0.97–0.99), having 0.94% sensitivity and 0.96% specificity, and an AUC of 0.99 (95% CI, 0.99–1.00), having 0.98% sensitivity and 1.00% specificity, respectively (Fig. 3A and 3B). As well, Fig. 3C and 3D show the ROC curves with an AUC of 0.81 (95% CI, 0.77–0.84), an AUC of 0.76 (95% CI, 0.72–0.81), in which ER and S were able to easily discriminate individuals with type 2 diabetes from type 1 diabetes with 1.00 (BPT), 5112.98 (mg^2/dL^2), as “optimal” cutoffs, having 0.66% sensitivity, 0.85% specificity, 0.68% sensitivity, and 0.72% specificity, respectively. Moreover, Fig. 3E and 3F show the ROC curves with an AUC of 0.72 (95% CI, 0.58–0.86), an AUC of 0.66 (95% CI, 0.47–0.86), in which ER and S were able to separate individuals with low risk (Negative Ab and 1 Ab) from high risk of rapid progression to stage 3 type 1 diabetes (≥ 2 Ab), with 0.52 (BPT), 923.65 (mg^2/dL^2), as “optimal” cutoffs, having 0.69% sensitivity, 0.74% specificity, 0.46% sensitivity, and 0.92% specificity, respectively. Therefore, the ROC curves and the corresponding AUCs show that both markers have a predictive ability for discrimination between the 3 different cases.

Discussion

In the present study, we define 2 different dynamical system approaches for evaluating the randomness, dynamics, and complexity of daily CGM data: (i) transitions through different glycemic states using a probability transition matrix and the ER criterion calculated from a symbolic encoding of the CGM profiles, and (ii) the area of the fitting ellipse calculated from 2-dimensional PP. These approaches can be used to inform decision support, and clinical and automated treatment algorithms. These dynamic markers can identify healthy individuals at heightened risk of progression to stage 3 type 1 diabetes. They also have the potential to track disease or treatment progression. Disease deterioration or progression is signified by transitions into undesired states and, conversely, a successful treatment optimization of medication titration is reflected by transition into clinically desirable states. Similarly, an increase in the S of the PP ellipse is a sign of increased variability and progression to stage 3 type 1 diabetes, or poor glycemic control.

The ability of both markers to distinguish with high fidelity between healthy and diabetic individuals could potentially serve as a CGM-based replacement for common diabetes diagnostic tests like OGTT—a 1-week CGM worn in the home environment, accompanied by a predefined schedule of meals and physical activity, is likely to achieve diagnostic results similar to those accepted in clinical practice, while greatly simplifying data collection and enabling telemedicine approaches (see Fig. 1). Additionally, it is the role of professional societies such as the American Diabetes Association and the European Association for the Study of Diabetes to establish standards for diagnosing diabetes. As future studies will no doubt be conducted, these new standards may merit inclusion in the list of methods for diagnosing diabetes. In particular, we found that the ER and the S of the PP ellipse as CGM dynamic markers can be indices for diagnosing diabetes, screening individuals at high risk of developing diabetes, and tracking disease progression over time. We found that the ER criterion

greater than 0.52 BPT and an area of the PP ellipse greater than 923.65 (mg^2/dL^2) are associated with a higher risk of progression to stage 3 type 1 diabetes as evaluated by the number of islet Abs. In addition, the ER criterion greater than 0.76 BPT, and an S of the PP ellipse greater than 1993.91 (mg^2/dL^2) are associated with the onset of diabetes. Fig. 2 shows an example of the transition probability matrix for an autoantibody-negative individual (eg, at low risk of developing type 1 diabetes), characterized by a low ER and a small S of the PP ellipse (see Fig. 2A and 2B) vs the transition probability matrix of an autoantibody-positive individual (with ≥ 2 Abs—at high immunological risk of developing type 1 diabetes), characterized by a high ER and a larger S of the PP ellipse (see Fig. 2C and 2D). Therefore, both CGM dynamics markers can be used as an alternative to, or in addition to, current screening and diagnostic methods/criteria.

Recently, TrialNet has demonstrated the heterogeneity in OGTT results within diabetes-staging categories with probable implications for diabetes progression, even with standardized testing procedures (41). Consequently, TrialNet in recent disease prevention trials has moved beyond classic risk categories to type 1 diabetes staging (3 states/stages). Whether CGM could be a viable alternative and replacement to the classical OGTT approach to detect imminent type 1 diabetes risk is debatable (42) and may largely depend on the way the CGM data is analyzed. Therefore, the proposed approaches could be a solution to detect dysglycemia and track the progression of the disease over time through the transition from one glycemic state to another with a specific time of staying in each, or through the area of the PP ellipse. A smaller, more concentrated plot indicates system (healthy individual) stability, whereas a more widespread PP indicates system (diabetic individual) irregularity, reflecting in our case poorer glucose control (unstable diabetes) or progression to the onset of diabetes.

Limitations of the present analysis include that the CGM data sets were collected using different versions of the Dexcom sensor (eg, Dexcom G4, G4 Platinum, G5, and G6), particularly evident in case 1 and case 2 of our analysis. However, our findings indicate that the CGM differences between different types/stages of diabetes are much more pronounced than between-sensor differences. Another limitation is that the healthy participant studies (TN and NoNDiab) have a small number of individuals and daily CGM profiles. Additionally, those individuals in the TN study were all relatives of individuals with type 1 diabetes probands. Besides that, there is no information about the islet autoantibody status for those individuals in the NoNDiab study.

In summary, the present study showed that the use of 1-week CGM data with both CGM dynamic markers, as new “mathematical” tools for diabetes diagnosis, could help improve diabetes risk detection; help identify early-stage type 1 diabetes before symptoms occur; and potentially alert individuals for early diagnosis or prevention. While gauging system dynamics requires certain calculations, the markers proposed here can be embedded in CGM output and visualization software without difficulty.

Acknowledgments

The authors would like to thank the UVA Center for Diabetes Technology Data Team for organizing the seven studies of

CGM and the TrialNet Coordinating Center for assistance with the recruitment of the Ab-positive participants.

Funding

This study was supported by a master research agreement between Dexcom, Inc. and the University of Virginia. Dexcom, Inc. placed no restrictions regarding submission of this article for publication. As well, this study was supported by the National Institutes of Health (NIH), National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) Award DP3DK106907 (TrialNet ancillary study), and Commonwealth Research Commercialization Fund (CRCF) Award MF20-007-LS. The Type 1 Diabetes TrialNet Study Group is a clinical trials network funded through a cooperative agreement by the National Institutes of Health (NIH) through the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), the National Institute of Allergy and Infectious Diseases (NIAID), and the Eunice Kennedy Shriver National Institute of Child Health and Human Development, and Juvenile Diabetes Research Foundation (JDRF).

Author Contributions

E.M. performed the analysis and verified the underlying data, developed the methodology, performed the computations, conducted the statistical analyses, and drafted the first version of the manuscript. E.M., L.F., and B.K. provided critical revision of the manuscript. B.K. designed the study and participated in the development of the analytical methodology. E.M., L.F., and B.K. were involved in the analyses and provided comments on drafts of the article. All authors reviewed and edited the manuscript. E.M. is the guarantor of this work and, as such, had full access to all the data in the study, and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Disclosures

E.M. has nothing to declare. L.F. declares research support from Dexcom and research support and patent royalties from Novo Nordisk, all managed by the University of Virginia. B.K. declares research support handled by the University of Virginia by Dexcom, Novo Nordisk, and Tandem Diabetes Care; and patent royalties handled by the University of Virginia by Dexcom, Johnson & Johnson, Novo Nordisk, and Sanofi.

Data Availability

Some or all data sets generated during and/or analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request.

References

- International Diabetes Federation. *IDF Diabetes Atlas*. 10th ed. International Diabetes Federation; 2021. Accessed June 5, 2024. <https://diabetesatlas.org/atlas/tenth-edition/>
- Harris MI, Eastman RC. Early detection of undiagnosed diabetes mellitus: a US perspective. *Diabetes Metab Res Rev*. 2000;16(4):230-236.
- ElSayed NA, Aleppo G, Aroda VR, *et al*. On behalf of the American diabetes association. 2. Classification and diagnosis of diabetes: standards of care in diabetes–2023. *Diabetes Care*. 2023;46(suppl 1):S19-S40.
- International Expert Committee. International expert committee report on the role of the A1C assay in the diagnosis of diabetes. *Diabetes Care*. 2009;32(7):1327-1334.
- World Health Organization. *Definition and Diagnosis of Diabetes Mellitus and Intermediate Hyperglycaemia: Report of a WHO/IDF Consultation*. World Health Organization; 2006. Accessed June 5, 2024. <https://apps.who.int/iris/handle/10665/43588>
- Echouffo-Tcheugui JB, Perreault L, Ji L, Dagogo-Jack S. Diagnosis and management of prediabetes: a review. *JAMA* 2023; 329(14):1206-1216.
- Harlass FE, Brady K, Read JA. Reproducibility of the oral glucose tolerance test in pregnancy. *Am J Obstet Gynecol*. 1991;164(2):564-568.
- Catalano PM, Avallone DA, Drago NM, Amini SB. Reproducibility of the oral glucose tolerance test in pregnant women. *Am J Obstet Gynecol*. 1993;169(4):874-881.
- Ko GT, Chan JC, Woo J, *et al*. The reproducibility and usefulness of the oral glucose tolerance test in screening for diabetes and other cardiovascular risk factors. *Ann Clin Biochem*. 1998;35(Pt 1):62-67.
- Klein KR, Walker CP, McFerrer AL, Huffman H, Frohlich F, Buse JB. Carbohydrate intake prior to oral glucose tolerance testing. *J Endocr Soc*. 2021;5(5):bvab049.
- Bergman M, Abdul-Ghani M, DeFronzo RA, *et al*. Review of methods for detecting glycemic disorders. *Diabetes Res Clin Pract*. 2020;165:108233.
- Dagogo-Jack S. Pitfalls in the use of HbA1(c) as a diagnostic test: the ethnic conundrum. *Nat Rev Endocrinol*. 2010;6(10):589-593.
- Ziemer DC, Kolm P, Weintraub WS, *et al*. Glucose-independent, black-white differences in hemoglobin A1c levels: a cross-sectional analysis of 2 studies. *Ann Intern Med*. 2010;152(12):770-777.
- Herman WH, Ma Y, Uwaifo G, *et al*. Differences in A1C by race and ethnicity among patients with impaired glucose tolerance in the diabetes prevention program. *Diabetes Care*. 2007;30(10):2453-2457.
- Khosla L, Bhat S, Fullington LA, Horlyck-Romanovsky MF. HbA1c performance in African descent populations in the United States with normal glucose tolerance, prediabetes, or diabetes: a scoping review. *Prev Chronic Dis*. 2021;18:E22.
- Kravarusic J, Aleppo G. Diabetes technology use in adults with type 1 and type 2 diabetes. *Endocrinol Metab Clin North Am*. 2020;49(1):37-55.
- Polonsky WH, Hessler D, Ruedy KJ, Beck RW; DIAMOND Study Group. The impact of continuous glucose monitoring on markers of quality of life in adults with type 1 diabetes: further findings from the DIAMOND randomized clinical trial. *Diabetes Care*. 2017;40(6):736-741.
- Rodbard D. Continuous glucose monitoring: a review of recent studies demonstrating improved glycemic outcomes. *Diabetes Technol Ther*. 2017;19(S3):S25-S37.
- Riddlesworth T, Price D, Cohen N, Beck RW. Hypoglycemic event frequency and the effect of continuous glucose monitoring in adults with type 1 diabetes using multiple daily insulin injections. *Diabetes Ther*. 2017;8(4):947-951.
- Kesavadev J, Vigersky R, Shin J, *et al*. Assessing the therapeutic utility of professional continuous glucose monitoring in type 2 diabetes across various therapies: a retrospective evaluation. *Adv Ther*. 2017;34(8):1918-1927.
- Montaser E, Fabris C, Kovatchev B. Essential continuous glucose monitoring metrics: the principal dimensions of glycemic control in diabetes. *Diabetes Technol Ther*. 2022;24(11):797-804.
- Helminen O, Pokka T, Tossavainen P, Ilonen J, Knip M, Veijola R. Continuous glucose monitoring and HbA1c in the evaluation of glucose metabolism in children at high risk for type 1 diabetes mellitus. *Diabetes Res Clin Pract*. 2016;120:89-96.
- Steck AK, Dong F, Taki I, Hoffman M, Klingensmith GJ, Rewers MJ. Early hyperglycemia detected by continuous glucose monitoring in children at risk for type 1 diabetes. *Diabetes Care*. 2014;37(7):2031-2033.

24. Steck AK, Dong F, Taki I, *et al.* Continuous glucose monitoring predicts progression to diabetes in autoantibody positive children. *J Clin Endocrinol Metab.* 2019;104(8):3337-3344.
25. Steck AK, Dong F, Geno Rasmussen C, *et al.* CGM metrics predict imminent progression to type 1 diabetes: autoimmunity screening for kids (ASK) study. *Diabetes Care.* 2022;45(2):365-371.
26. Wilson DM, Pietropaolo SL, Acevedo-Calado M, *et al.* CGM metrics identify dysglycemic states in participants from the TrialNet pathway to prevention study. *Diabetes Care* 2023;46(3):526-534.
27. Montaser E, Brown SA, DeBoer MD, Farhy LS. Predicting the risk of developing type 1 diabetes using a one-week continuous glucose monitoring home test with classification enhanced by machine-learning: an exploratory study. *J Diabetes Sci Technol.* 2023;18(2):257-265.
28. Kovatchev BP, Lobo B. Clinically similar clusters of daily continuous glucose monitoring profiles: tracking the progression of glycemic control over time. *Diabetes Technol Ther.* 2023;25(8):519-528.
29. Kovatchev B, Anderson SM, Raghinaru D, *et al.* Randomized controlled trial of mobile closed-loop control. *Diabetes Care.* 2020;43(3):607-615.
30. Brown SA, Kovatchev BP, Raghinaru D, *et al.* Six-month randomized, multicenter trial of closed-loop control in type 1 diabetes. *N Engl J Med.* 2019;381(18):1707-1717.
31. Breton MD, Kanapka LG, Beck RW, *et al.* A randomized trial of closed-loop control in children with type 1 diabetes. *N Engl J Med.* 2020;383(9):836-845.
32. Beck RW, Riddlesworth T, Ruedy K, *et al.* Effect of continuous glucose monitoring on glycemic control in adults with type 1 diabetes using insulin injections: the DIAMOND randomized clinical trial. *JAMA.* 2017;317(4):371-378.
33. Beck RW, Riddlesworth TD, Ruedy K, *et al.* Continuous glucose monitoring versus usual care in patients with type 2 diabetes receiving multiple daily insulin injections: a randomized trial. *Ann Intern Med.* 2017;167(6):365-374.
34. Shah VN, DuBose SN, Li Z, *et al.* Continuous glucose monitoring profiles in healthy nondiabetic participants: a multicenter prospective study. *J Clin Endocrinol Metab.* 2019;104(10):4356-4364.
35. Montaser E, Breton M, Brown S, DeBoer M, Kovatchev B, Farhy L. Predicting immunological risk for stage 1 and stage 2 diabetes using a 1-week CGM home test, nocturnal glucose increments, and standardized liquid mixed meal breakfasts, with classification enhanced by machine learning. *Diabetes Technol Ther* 2023;25(9):631-642.
36. Montaser E, Farhy L, Kovatchev B. Supplemental materials for "Novel Detection and Progression Markers for Diabetes Based on Continuous Glucose Monitoring Data Dynamics". *Figureshare.* 2024. <https://figshare.com/s/714281fc133dc64eb245>. Date of deposit 21 May 2024.
37. Thomas MC, Joy AT. *Elements of Information Theory.* 2nd ed. Wiley; 2006.
38. Kovatchev BP, Clarke WL, Breton M, Brayman K, McCall A. Quantifying temporal glucose variability in diabetes via continuous glucose monitoring: mathematical methods and clinical application. *Diabetes Technol Ther.* 2005;7(6):849-862.
39. Breton MD, Shields DP, Kovatchev BP. Optimum subcutaneous glucose sampling and Fourier analysis of continuous glucose monitors. *J Diabetes Sci Technol.* 2008;2(3):495-500.
40. Brennan M, Palaniswami M, Kamen P. Do existing measures of poincaré plot geometry reflect nonlinear features of heart rate variability? *IEEE Trans Biomed Eng.* 2001;48(11):1342-1347.
41. Krischer JP, Schatz DA, Bundy B, Skyler JS, Greenbaum CJ. Effect of oral insulin on prevention of diabetes in relatives of patients with type 1 diabetes: a randomized clinical trial. *JAMA.* 2017;318(19):1891-1902.
42. Ylescupidez A, Speake C, Pietropaolo SL, *et al.* OGTT metrics surpass continuous glucose monitoring data for T1D prediction in multiple-autoantibody-positive individuals. *J Clin Endocrinol Metab.* 2023;109(1):57-67.